

Clinical Risk Assessment Report: Patient 71

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 14.8% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.67x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT), which reduced the patient's absolute risk by 1.4%. Biologically, this feature quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Patient Age (Age) **Value: 53.0 [Bottom 16%] (+2.3%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.83 [Top 20%] (+1.0%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

3. Protective / Mitigating Factors (Reducing Risk)

Tumor Sphericity (CT) (SHAPE_Sphericity(onlyFor3DROI)).CT) **Value: 0.77 [Top 21%] (-1.4%)**
Quantifies how closely the tumor resembles a perfect sphere; lower values indicate irregular, spiculated, or highly invasive margins.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: 0.33 [Top 11%] (-1.4%)**
Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.03 [Avg (46th %ile)] (-1.2%)**
Reflects the most dense solid components or calcifications within the primary tumor lesion.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: 0.1 [Top 13%] (-0.9%)**
Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.8 [Avg (38th %ile)] (-0.7%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

Large Zone Low Gray-Level Emphasis (CT) (GLZLM_LZLGE.CT) **Value: -0.1 [Top 18%] (-0.6%)**
Indicates the dominance of large regions with low attenuation or metabolism, pointing to extensive necrosis or ground-glass opacities.

Machine Learning Feature Attribution (SHAP Decision Path)

